# A 12-VERTEX TREE REFUTING THE $GRM_{-2}$ LOWER BOUND FOR TREES OF MAXIMUM DEGREE FOUR

Abstract. For a graph $G$ and $\lambda \in \mathbb{R}$, the general reduced second Zagreb index is $GRM_\lambda(G) = \sum_{uv\in E(G)}(\deg u + \lambda)(\deg v + \lambda)$. Bašić and Ilić assert that every tree $T$ of order $n$ with $\Delta(T) = 4$ and $n \not\equiv \{1, 2, 3\} \pmod 4$ satisfies $GRM_{-2}(T) \geq -n$, and that this value is attained. We exhibit a tree on 12 vertices with maximum degree 4 and $GRM_{-2} = -13 < -12$, so the assertion is false, and we exhibit an infinite family showing that for every $n \equiv 0 \pmod 4$ with $n \geq 16$ the assertion fails by 2. Consequently the claimed extremal family $TT^4_{opt}(k)$ is not extremal for any $k \geq 2$. The refutation is a finite hand computation on eleven edges. We do not determine the exact minimum at any order, we do not decide whether the assertion holds at $n = 8$, and we make no claim about the source's other results.

## 1. The claim under test, and its locator

For a graph $G$ and a real parameter $\lambda$, Horoldagva, Buyantogtokh, Das and Lee [3] introduced the *general reduced second Zagreb index*

$$GRM_\lambda(G) = \sum_{uv\in E(G)} \big(\deg(u) + \lambda\big)\big(\deg(v) + \lambda\big), \tag{1}$$

so that

$$GRM_{-2}(T) = \sum_{uv\in E(T)} \big(\deg(u) - 2\big)\big(\deg(v) - 2\big). \tag{2}$$

Write $m_{ij}$ for the number of edges of $T$ whose endpoint degrees are $i$ and $j$, and $n_i$ for the number of vertices of degree $i$.

Bašić and Ilić [1] devote their Section 4 to the minimum of $GRM_{-2}$ over trees of maximum degree 4. All references below are to line numbers of the arXiv LaTeX e-print of [1], version 1; the class is fixed at line 797 by the phrase "*for any tree $T$ with a maximal degree of $\Delta = 4$*", and the section heading at line 766 reads "*Minimum value of $GRM_{-2}$ in the class of molecular trees with maximal degree 4*". The source's Section 4 rests on its own reduction, the displayed equation at line 800,

$$-GRM_{-2}(T) = -3m_{12} - m_{13} - m_{22} - m_{34} - 3m_{44} + n - n_3 + 3, \tag{3}$$

2020 *Mathematics Subject Classification.* Primary 05C09; Secondary 05C05, 05C07, 05C35, 92E10.

*Key words and phrases.* general reduced second Zagreb index, $GRM_\lambda$, reduced Zagreb index, molecular tree, maximum degree, extremal tree, counterexample.

which in turn comes from its pre-substitution formula at line 795,

$$-GRM_{-2}(T) = m_{13} + 2m_{14} - m_{33} - 2m_{34} - 4m_{44}. \tag{4}$$

The section closes four residue cases in turn. The last of them, at line 838, is quoted here character for character:

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Therefore, based on equation (\ref{main2}), if $n
\not\equiv \{1, 2, 3\} \pmod{4}$, it follows that
$GRM_{-2}(T) \geq -n$.
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With the class of line 797, this is the following statement, which we refute.

> **Claim** ([1], line 838). *For every tree $T$ of order $n$ with $\Delta(T) = 4$ and $n \equiv 0 \pmod 4$ one has $GRM_{-2}(T) \geq -n$.*

Line 842 adds that the value $-n = -(4k+4)$ *is attained*, at parameters listed at lines 843 and 844, and line 845 names the family $TT^4_{opt}(k)$ of trees realising it. The adjacent improvement of Dehgardi and Klavžar [2], reference [3] of [1], is restricted to $\lambda \geq -\frac{1}{2}$ and so says nothing about $\lambda = -2$.

## 2. The counterexample

**Theorem 1.** *There is a tree $W_1$ on $n = 12$ vertices with $\Delta(W_1) = 4$ and $GRM_{-2}(W_1) = -13$. Since $12 \equiv 0 \pmod 4$ and $-13 < -12 = -n$, the Claim of Section 1 is false.*

*Proof.* Let $W_1$ have vertex set $\{0, 1, \ldots, 11\}$ and the eleven edges

$$01, \quad 02, \quad 03, \quad 04, \quad 15, \quad 56, \quad 57, \quad 68, \quad 89, \quad 8\text{-}10, \quad 8\text{-}11. \tag{5}$$

Pictorially, 0 and 8 are the two branch vertices, joined by the path $0, 1, 5, 6, 8$, with a single extra leaf at 5:

$$\begin{array}{ccccccccc}
2\ 3\ 4 & & & & & & & & 9\ 10\ 11 \\
\diagdown | \diagup & & & & & & & & \diagdown | \diagup \\
0 & — & 1 & — & 5 & — & 6 & — & 8 \\
 & & & & | & & & & \\
 & & & & 7 & & & &
\end{array}$$

*$W_1$ is a tree.* It has 12 vertices and $11 = n - 1$ edges, no loop and no repeated edge in (5), and every vertex is reached from 0 by $0 \to \{1, 2, 3, 4\}$, $1 \to 5$, $5 \to \{6, 7\}$, $6 \to 8$, $8 \to \{9, 10, 11\}$. A connected graph with $n - 1$ edges is a tree.

*Degrees.* From (5), $\deg 0 = \deg 8 = 4$, $\deg 5 = 3$, $\deg 1 = \deg 6 = 2$, and $2, 3, 4, 7, 9, 10, 11$ are leaves. The degree sequence is $(4, 4, 3, 2, 2, 1^7)$, whose sum is $22 = 2 \cdot 11$. In particular $\Delta(W_1) = 4$ exactly, attained at 0 and at 8, so $W_1$ lies in the class of line 797 under either reading of it, $\Delta = 4$ or $\Delta \leq 4$.

*The value.* Evaluate (2) edge by edge. Only the seven edges incident with a leaf contribute, because an edge with an endpoint of degree 2 contributes

0.

| edges | end degrees | each | subtotal |
|---|---|---|---|
| 02, 03, 04 | $(4,1)$ | $(4-2)(1-2)=-2$ | $-6$ |
| 89, 8-10, 8-11 | $(4,1)$ | $-2$ | $-6$ |
| 57 | $(3,1)$ | $(3-2)(1-2)=-1$ | $-1$ |
| 01 | $(4,2)$ | $(4-2)(2-2)=0$ | 0 |
| 15 | $(2,3)$ | 0 | 0 |
| 56 | $(3,2)$ | 0 | 0 |
| 68 | $(2,4)$ | 0 | 0 |
| | | total | $-13$ |

Hence $GRM_{-2}(W_1) = -13$, while the Claim asserts $GRM_{-2}(W_1) \geq -12$. □

Two remarks fix the scope of Theorem 1. First, $W_1$ itself violates only the Claim of line 838: the source's earlier conclusions at line 822 ($GRM_{-2} \geq -(n+2)$ for $n \neq 4k+1$) and line 830 ($GRM_{-2} \geq -(n+1)$ for $n \not\equiv \{1,2\} \pmod 4$) both survive at $W_1$, the second with equality, since $-13 = -(n+1)$. The tree $W_2$ of Section 3 violates line 830 as well. Second, at $n = 8$ the bound $-n$ is attained, by the double star obtained by joining two vertices of degree 4, which has $m_{14} = 6$, $m_{44} = 1$ and $GRM_{-2} = 6 \cdot (-2) + 1 \cdot 4 = -8 = -n$; we do not decide whether $GRM_{-2}(T) \geq -8$ holds for every tree $T$ of order 8 with $\Delta(T) = 4$, and nothing here depends on that.

The edge partition of $W_1$ is $m_{13} = 1$, $m_{14} = 6$, $m_{23} = 2$, $m_{24} = 2$, all other $m_{ij} = 0$, with $n_1 = 7$, $n_2 = 2$, $n_3 = 1$, $n_4 = 2$; both (3) and (4) return $-GRM_{-2}(W_1) = 13$.

## 3. An infinite family

**Proposition 2.** *For every $k \geq 3$ there is a tree $F(k)$ of order $n = 4k+4$ with $\Delta = 4$ and $GRM_{-2}(F(k)) = -(n+2)$.*

*Proof.* Take any tree $S$ (a *skeleton*) on $k+1$ nodes in which a distinguished node $v$ has degree 3 and every node has degree at most 4; such an $S$ exists for every $k \geq 3$, for instance the tree on $\{0,1,\ldots,k\}$ with edges $01, 02, 03$ together with the path edges $i(i+1)$ for $3 \leq i \leq k-1$ (no path edge at all when $k = 3$, where $S = K_{1,3}$), whose node $v = 0$ has degree 3. Subdivide every edge of $S$ exactly once, which creates $k$ new vertices of degree 2, and then attach $4 - d_S(u)$ pendant vertices to each node $u \neq v$, where $d_S(u)$ is the degree of $u$ in $S$.

In the resulting tree $F(k)$ the node $v$ has degree 3 and all three of its neighbours are subdivision vertices, so $n_3 = 1$ and $m_{13} = 0$; each node $u \neq v$ has degree 4, so $n_4 = k$; the $k$ subdivision vertices give $n_2 = k$; and every edge of $F(k)$ either joins a degree-2 vertex to something or joins a degree-4 vertex to a leaf, whence $m_{12} = m_{22} = m_{33} = m_{34} = m_{44} = 0$. The skeleton degree sum is $2k$, so $\sum_{u \neq v} d_S(u) = 2k-3$ and the number of pendant vertices is $n_1 = m_{14} = 4k - (2k-3) = 2k+3$. Therefore

$$n = n_1 + n_2 + n_3 + n_4 = (2k+3) + k + 1 + k = 4k + 4,$$

and by (2) only the $m_{14}$ edges contribute, each $-2$, so $GRM_{-2}(F(k)) = -2(2k+3) = -(4k+6) = -(n+2)$. $\square$

For $k = 3$ this is the tree $W_2$ on 16 vertices with edges

01, 02, 03, 04, 15, 56, 57, 68, 79, 8-10, 8-11, 8-14, 9-12, 9-13, 9-15,

with $\Delta = 4$ attained at $0, 8, 9$, $m_{14} = 9$ and $GRM_{-2}(W_2) = -18 < -16$. Note that $W_2$ violates the source's line 830 as well, since $-18 < -17 = -(n+1)$, while $W_1$ does not.

**Corollary 3.** *The family $TT^4_{opt}(k)$ of* [1]*, line 845, is not the set of extremal trees of order $n = 4k+4$ for any $k \geq 2$.*

*Proof.* Both parameter sets the source assigns to its optimal tree have $m_{13} = m_{33} = m_{34} = 0$; the first (line 843) has $m_{14} = 2k+2$ and $m_{44} = 0$, the second (line 844) has $m_{14} = 2k+4$ and $m_{44} = 1$. By (4) they give $-GRM_{-2} = 2(2k+2) = 4k+4 = n$ and $-GRM_{-2} = 2(2k+4) - 4 = 4k+4 = n$ respectively, i.e. $GRM_{-2} = -n$ in both cases. By Theorem 1 and Proposition 2 the minimum at $n = 4k+4$ is at most $-(n+1)$ for $k = 2$ and at most $-(n+2)$ for $k \geq 3$, strictly below $-n$. $\square$

## 4. What is not settled

- *No exact minimum is claimed.* Theorem 1 and Proposition 2 give upper bounds on $\min\{GRM_{-2}(T) : |V(T)| = n,\ \Delta(T) = 4\}$ only. We do not prove a matching lower bound at any order, we do not claim that the trees exhibited here are minimisers, and we make no claim about the number of minimisers.
- *No corrected replacement for $TT^4_{opt}(k)$ is proved here.* The inductive structural characterisation at line 846 of [1] is not repaired.
- *The disputed statements are line 838, the line 830 bound at $n \equiv 0 \pmod 4$ with $n \geq 16$, and the attainment claim at lines 842–846.* We assert nothing about the source's $\Delta = 3$ results, its $\lambda \geq -1$ material, or its $\Delta = 4$ conclusions at $n \equiv 1, 2, 3 \pmod 4$.
- *"Molecular tree" is never separately defined* in [1]; we read the class as it is fixed at line 797. All trees exhibited here have $\Delta$ exactly 4, so the refutation is insensitive to reading the class as $\Delta \leq 4$ instead.
- *What the accompanying program does.* The recorded run in the folder of this note recomputes $GRM_{-2}$ and the degree and edge counts of $W_1$, $W_2$, the tree $F(4)$ on 20 vertices and the $n = 8$ double star from their edge lists, and instantiates $F(k)$ for $k = 3, \ldots, 12$. It enumerates no trees, so it decides no minimum and does not address $n = 8$ beyond the double star; the general statement of Proposition 2 for all $k \geq 3$ is established by the proof above, not by that run.

- *This corrects a preprint.* [1] has one version, no journal reference and no erratum at the time of writing. Should a refereed version appear with a repaired Section 4, the target of this note would cease to exist in the literature, though nothing about $W_1$ would change.

## References

[1] M. Bašić and A. Ilić, *Further results on the lower bound on reduced Zagreb index of trees*, arXiv:2604.06044, 2026. (All line numbers cited here refer to `main.tex` of the version-1 e-print, 966 lines, 61,112 bytes.)

[2] N. Dehgardi and S. Klavžar, *Improved lower bounds on the general reduced second Zagreb index of trees*, preprint, 2023; cited as reference [3] of [1]. Published form: N. Dehgardi, S. Klavžar and M. Azari, J. Comb. Optim. **51** (2026), Paper No. 50 (arXiv:2605.30402); it treats $\lambda \geq -\frac{1}{2}$ only.

[3] B. Horoldagva, L. Buyantogtokh, K. C. Das and S.-G. Lee, *On general reduced second Zagreb index of graphs*, Hacettepe J. Math. Stat. **48** (2019), 1046–1056.